

(AnnexureB)

PROJECT REPORT

On

BookMyHaven

22CS422

of

**COMPUTER SCIENCE AND ENGINEERING
B.E. Batch-2022 (8th Semester)**



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CERTIFICATE

This is to be certified that the project entitled “Doctor Management system ” has been submitted for the Bachelor of Computer Science Engineering at Chitkara University, Punjab during the academic semester jan 2026-july 2026 is a bonafide piece of project work carried out by “Lokesh(2210991855), Lakshya Puri(2210991845) , Manthan Gupta(2210991894), Makul Malhotra(2210991867) towards the partial fulfillment for the award of the course IOHEunder the guidance of “Dr. Preeti Saini” and supervision.

Sign. of Project Guide

Doctor Management System

Dr. Preeti Saini

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CANDIDATE’SDECLARATION

We, Students of the Chitkara University, Punjab hereby declare that the IOHE Report entitled “**BookMyHaven**” is an original work and data provided in the study is authentic to the best of our knowledge. This report has not been submitted to any other Institute for the award of any other course.

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ACKNOWLEDGEMENT

It is our pleasure to be indebted to various people, who directly or indirectly contributed in the development of this work and who influenced my thinking, behavior and acts during the course of study.

We express our sincere gratitude to all for providing mean opportunity to undergo IOHE as the part of the curriculum.

We are thankful to Dr .Preeti Saini for his support, cooperation, and motivation provided to us during the training for constant inspiration, presence and blessings.

We also extend our sincere appreciation to **Dr.Preeti Saini** Who provided his valuable suggestions and precious time in accomplishing our IOHE report.

Lastly, we would like to thank the almighty and our parents for their moral support and friends with whom we share dour day-to day experience and received lots of suggestions that improve our quality of work.

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Abstract

BookMyHaven is a web-based hotel booking platform developed to simplify and modernize the process of searching, comparing, and reserving accommodations online. The system provides users with a clean and user-friendly interface where they can search hotels based on location, travel dates, and guest details. The platform includes advanced filtering options such as price range, star ratings, and amenities, helping users find hotels according to their preferences. Detailed hotel information including pricing, images, ratings, and available facilities is displayed to improve decision-making. BookMyHaven also provides secure user authentication for login and signup functionality. The project is developed using modern technologies such as React.js, Node.js, Express.js, and MongoDB to ensure scalability, efficiency, and maintainability. The main objective of this project is to improve the traditional hotel booking process by providing a centralized, efficient, and reliable digital solution.

Keywords

- Hotel Booking System
- React.js
- Node.js
- MongoDB
- Web Application
- Authentication
- Hotel Reservation
- Booking Platform

2. Introduction to the Project

BookMyHaven is an online hotel booking platform designed to help users efficiently search, compare, and reserve accommodations. The application simplifies the hotel booking process by offering real-time search functionality, detailed hotel listings, secure authentication, and filtering options. The system allows users to explore available hotels based on location, check-in/check-out dates, and guest details. Users can also filter hotels according to price, ratings, and amenities to find accommodations that meet their requirements. The platform focuses on usability, transparency, and efficiency, making the hotel booking process faster and more convenient.

2.1 Background

Traditional hotel booking methods often involve manual searching, lack of centralized information, and inefficient comparison of hotels. Many online booking systems are overly complex and require unnecessary registration before users can browse hotels. Additionally, users often face difficulties in finding hotels that match their budget, preferred location, and required facilities. These challenges create a need for a simple, user-friendly, and efficient hotel reservation platform. BookMyHaven was developed to overcome these limitations and provide users with a seamless digital hotel booking experience.

2.2 Problem Statement

Users face several challenges while booking hotels through traditional or existing online systems, including:

- Lack of centralized hotel information
- Complex and confusing user interfaces
- Limited filtering and comparison options
- Difficulty in finding hotels based on budget and preferences
- Lack of real-time hotel availability information
- Time-consuming booking process

Therefore, there is a need for a modern web-based hotel booking system that provides easy navigation, advanced

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filtering, and secure booking functionality.

3. Software and Hardware Requirement Specification

Software Requirements

Software	Purpose
React.js	Frontend development
Node.js	Backend runtime
Express.js	Server-side framework
MongoDB	Database
VS Code	Development environment
GitHub	Version control
Vite	Frontend build tool

Hardware Requirements

Hardware	Specification
Processor	Intel i3 or above
RAM	Minimum 4GB
Storage	Minimum 20GB
Internet	Required for deployment and APIs

3.1 Methods

The project was developed using a modular and component-based development approach. The frontend and backend were implemented separately and connected using APIs.

The development process included:

- Requirement analysis
- System design
- Frontend and backend development
- Database integration
- Testing and debugging
- Deployment preparation

3.2 Programming / Working Environment

The application was developed in the following working environment:

- Operating System: Windows 10/11
- IDE: Visual Studio Code
- Frontend Framework: React.js with Vite
- Backend Environment: Node.js and Express.js
- Database: MongoDB
- Version Control: Git and GitHub

3.3 Requirements to Run the Application

To run the application successfully, the following requirements are necessary:

- Node.js installed
- MongoDB installed or connected through cloud database
- Modern web browser (Chrome, Edge, Firefox)
- Internet connection
- Required npm packages installed using:
npm install

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To start the application:

```
npm run dev
```

4. Database Analyzing, Design and Implementation

MongoDB was used as the database management system for storing user and hotel information.

The database contains collections such as:

- Users
- Hotels
- Bookings
- Authentication Data

The database design ensures efficient storage and retrieval of hotel details, booking information, and user credentials. Backend APIs interact with the database to fetch, update, and manage records dynamically.

5. Program's Structure Analyzing and GUI Constructing

The application follows a modular structure where different functionalities are divided into separate components and modules.

Main Modules

- Homepage Module
- Authentication Module
- Hotel Search Module
- Filtering Module
- Hotel Listing Module
- Database Management Module

GUI Construction

The graphical user interface was designed using React.js and CSS/Tailwind CSS. The interface focuses on:

- Responsive design
- Easy navigation
- Attractive layout
- User-friendly interactions

The homepage includes a search panel, hotel listings, and filtering options to enhance user experience.

6. Code-Implementation and Database Connections

The frontend was implemented using React.js components, while backend functionality was developed using Node.js and Express.js.

MongoDB was connected to the backend using database connection libraries to store and retrieve data efficiently.

APIs were created for:

- User authentication
- Hotel searching
- Hotel listing retrieval
- Filtering operations
- Booking management

The frontend communicates with backend APIs to display dynamic hotel data and user information.

7. System Testing

Testing was performed to ensure proper functionality and performance of the application.

Testing Performed

- Login and Signup Testing
- Hotel Search Testing
- Filtering Functionality Testing
- Database Connectivity Testing
- Responsive UI Testing

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- Navigation Testing

The testing process helped identify and resolve bugs to improve application performance and user experience.

8. Limitations

Although BookMyHaven provides efficient hotel booking functionality, some limitations still exist:

- No integrated online payment gateway
- Limited real-time booking synchronization
- No AI-based recommendation system
- Limited admin functionalities
- No mobile application support currently

These limitations can be improved in future versions of the project.

9. Conclusion

BookMyHaven successfully provides a modern and user-friendly hotel booking platform that simplifies hotel searching and reservation processes.

The project demonstrates the use of modern web technologies such as React.js, Node.js, Express.js, and MongoDB to create a scalable and responsive web application. The platform improves user experience by providing efficient search, filtering, authentication, and hotel listing functionalities.

The system overcomes many traditional booking challenges and provides a reliable solution for online hotel reservations.

10. Future Scope

The project can be further enhanced by implementing advanced features such as:

- Online payment gateway integration
- AI-based hotel recommendation system
- Real-time booking confirmation
- Mobile application development
- Customer reviews and ratings
- GPS-based nearby hotel suggestions
- Admin dashboard and analytics system

These features can make BookMyHaven more intelligent, efficient, and suitable for large-scale commercial deployment.

11. Bibliography / References

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